

Iterative Receiver and Code Design for Protograph-Coded Overloaded SCMA Systems

Yunye Li¹, Zhaojie Yang², Yong Liang Guan², *Senior Member, IEEE*, and Yi Fang³, *Senior Member, IEEE*

Abstract—This work investigates iterative receiver and code design for protograph-coded overloaded sparse code multiple access (SCMA) systems. As a type of non-orthogonal multiple access (NOMA) technology, SCMA relies on an iterative detection and decoding (IDD) receiver to eliminate inter-user interference and channel noise. However, the existing IDD receiver requires a larger number of iterations to achieve successful convergence. To overcome this limitation, we propose a *ranked successive detection and decoding (R-SDD)* receiver, which dynamically allocates computational resources to bottleneck users with the lowest reliability. This approach accelerates information convergence and improves error performance without increasing overall complexity. Moreover, we introduce a *factor-graph-based (FG)* interleaving scheme for the proposed SCMA systems, which outperforms the existing interleaver. To further enhance the error-correction capability for overloaded multi-user scenarios, we propose a new protograph design principle tailored for SCMA systems and develop two high-performance protograph low-density parity-check (LDPC) codes, achieving over 0.3 dB improvement compared with the state-of-the-art alternatives. Owing to the salient advantages of the new iterative receiver and code design, the proposed SCMA system offers a promising solution for future high-bandwidth, high-reliability communication networks.

Index Terms—Sparse code multiple access, iterative receiver, protograph codes, multi-user communication.

I. INTRODUCTION

WITH the proliferation of communication devices, the demand for higher bandwidth and massive connectivity is rapidly increasing, especially in Internet of Things (IoT) scenarios. Orthogonal multiple access (OMA) technologies, such as orthogonal frequency division multiplexing (OFDM) and code division multiple access (CDMA), have been widely adopted to improve spectral efficiency and support multi-user access [1]. However, existing studies indicate that non-orthogonal multiple access (NOMA) has the potential to

achieve larger capacity [2], [3], driving increased interest with respect to traditional OMA techniques. In 2013, Nikopour et al. proposed sparse code multiple access (SCMA) [4] as a new branch of NOMA. SCMA has demonstrated superior performance compared with other NOMA schemes, such as power-domain NOMA (PD-NOMA), low-density spreading (LDS) and pattern division multiple access (PDMA) [5], [6]. As a result, SCMA has attracted more attention in recent years and appeared as a promising candidate for 6G multiple access solutions.

SCMA introduces inter-user interference, which significantly deteriorates the system performance. Traditional OMA systems do not encounter such issue. To address this problem, an iterative detection and decoding (IDD) receiver was first introduced to SCMA systems in [7]. The IDD structure enables the detector to utilize soft information output from the decoder, thereby more effectively eliminating inter-user interference. To further improve system performance, low-density parity-check (LDPC) codes were integrated with an IDD structure that leverages extrinsic information in SCMA systems [8], leading to significant performance improvements and establishing a foundational framework for current SCMA research.

As a matter of fact, the performance of SCMA systems heavily relies on the IDD receiver [9]. As a comparison, in other iterative systems such as bit-interleaved coded modulation with iterative decoding (BICM-ID), IDD only serves as an enhancement [10], [11]. In contrast, SCMA systems possess poor bit error rate (BER) performance without IDD due to the severe inter-user interference. Meanwhile, IDD receiver introduces high complexity to the system. For this reason, improving system performance within a limited complexity becomes a challenging problem, which constitutes one of the primary motivations for this paper.

In the past decade, a great deal of research effort has been devoted to designing efficient IDD receivers for SCMA systems. The successive interference cancellation (SIC) structure has been employed in SCMA systems to resolve codebook collisions in grant-free scenarios [12]. In [13], an IDD receiver based on the minimum mean-square error with parallel interference cancellation (MMSE-PIC) algorithm was proposed, which achieves a fair performance at relatively low complexity. Among all detection algorithms for SCMA, the message-passing algorithm (MPA) is the most widely adopted due to its near-optimal performance compared with the maximum likelihood (ML) algorithm [14]. However, MPA also suffers from high computational complexity. To address this challenge,

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various modifications of the MPA have been explored [15], [16]. Moreover, a low-complexity message passing algorithm has been proposed in [17], which minimizes the variational free energy associated with the global *a-posteriori* probability.

Designing capacity-approaching codes for NOMA systems is a popular topic since they can significantly enhance error-correction capabilities. As a type of structured LDPC codes, protograph codes can achieve superior performance and lower complexity compared with traditional irregular LDPC codes [18]. However, in overloaded SCMA systems, the behavior of codes diverges significantly from that in single-user scenarios. Therefore, it is essential to design multi-user codes tailored for SCMA systems. In recent years, several multi-user codes with excellent performance have been proposed. In particular, LDPC codes tailored for the two-user Gaussian multiple access channel (GMAC) were designed through extrinsic information transfer (EXIT) analysis in [19]. Moreover, approximate message passing (AMP) algorithm was proposed to design LDPC codes for the large random matrix system (LRMS) [20]. The AMP algorithm can also provide valuable guidance for code design in other NOMA scenarios [21]. In addition, [22] explored the design of protograph code in the GMAC scenario.

However, to the best of our knowledge, there has been no protograph-code design specifically tailored for SCMA systems so far. Most of the existing multi-user codes are designed based on two-user scenario, which limits the overall performance in overloaded SCMA systems. Additionally, the code design principle which have been proposed for LRMS or GMAC [20], [22] may not suit for SCMA. Motivated by such issue, we investigate the code design tailored for SCMA systems and propose new protograph codes with desirable performance.

The main drawback of the traditional IDD structure is that the decoders for multiple users operate in a parallel manner. This structure prevents users from sharing decoding output within one outer iteration. To resolve this issue, we propose a novel IDD structure, referred to as *ranked successive detection and decoding (R-SDD)*. In the proposed structure, the decoders are sequentially connected, allowing each subsequent decoder to benefit from the decoding outputs of the preceding ones. Furthermore, the mutual information is used to determine the decoding order, optimizing the overall performance of the R-SDD receiver. Finally, we design high-performance protograph codes tailored for the proposed SCMA system with R-SDD. More specifically, the key contributions of our work are summarized as follows:

- 1) We propose an R-SDD receiver for protograph-coded SCMA systems. Compared with the traditional IDD structure, our proposed R-SDD receiver shows significant advantages in terms of both BER performance and complexity. Moreover, the trade-off between performance and complexity can be more flexibly managed through our design.
- 2) We develop a new factor-graph-based (FG) interleaving scheme to realize the appropriate mapping between the variable nodes and the SCMA symbols, which can achieve desirable iterative gain and hence improve the error performance of protograph-coded SCMA systems.

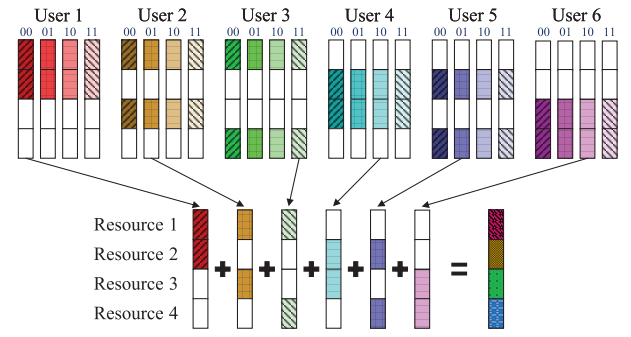


Fig. 1. Example of a 4×6 SCMA codebook.

- 3) We demonstrate that existing LDPC codes are unsuitable for the iterative receivers employed in SCMA systems. To overcome this limitation, we analyze the iterative convergence behavior of protograph codes with the aid of modified P-EXIT algorithm and establish a new design principle for protograph codes in multi-user scenarios. Based on the principle, we propose two high-performance protograph codes tailored for SCMA systems, offering significant performance improvement.
- 4) We conduct theoretical analysis and BER simulations to confirm that our proposed SCMA system, including the R-SDD receiver, FG interleaver, and newly designed protograph codes, achieves notable performance improvements compared with existing benchmarks over additive white Gaussian noise (AWGN) channel, Rayleigh fading channel, and multipath fading channel.

The remainder of this paper is organized as follows. In Section II, we review the SCMA system with traditional IDD receiver. Section III elaborates the proposed R-SDD receiver with FG interleaving scheme. In section IV, we discuss the design principle of protograph codes in SCMA systems and propose two high-performance multi-user codes. Section V analyzes the computational complexity of our R-SDD receiver in comparison with the traditional IDD structure. In Section VI, we evaluate the performance of the proposed SCMA system under various conditions, and finally Section VII concludes the paper.

II. SCMA SYSTEM WITH TRADITIONAL IDD RECEIVER

A. System Model

The structure of an SCMA symbol is defined by an SCMA codebook, as shown in Fig. 1. The number of users J is usually greater than the number of resources K , making SCMA an overloaded system. This enables higher throughput compared with an OMA system. The resource allocation among users is usually described by a factor graph matrix. For example, based on Fig. 1, the factor graph matrix with $K = 4$ and $J = 6$ is given by:

$$\mathbf{F}_{4 \times 6} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}. \quad (1)$$

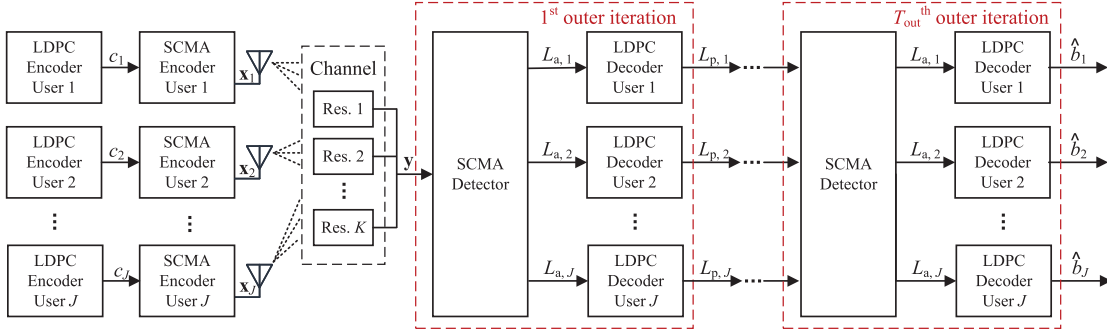


Fig. 2. Block diagram of an LDPC-coded SCMA system with traditional IDD receiver.

where the i -th row and j -th column represent the resource node (RN) and the user node (UN), respectively, the (k, j) -th element $f_{k,j} \in \mathbf{F}_{4 \times 6}$ indicates whether the k -th resource carries the message of user j , where $f_{k,j} = \{0, 1\}$.

To generate and decode SCMA symbols, the SCMA encoder and decoder are required. The block diagram of an LDPC-coded SCMA system is shown in Fig. 2. At the transmitter, for each user j ($j = 1, 2, \dots, J$), the information bit sequence $\mathbf{b}_j = [b_j^1, b_j^2, \dots, b_j^l]^T$ is encoded into the codeword $\mathbf{c}_j = [c_j^1, c_j^2, \dots, c_j^n]^T$ by the LDPC encoder, where l and n represent information bit length and codeword length, respectively. Subsequently, every $\log_2 M$ encoded bits $[c_j^1, c_j^2, \dots, c_j^{\log_2 M}]^T$ are grouped and mapped to a K -dimensional modulated symbol $\mathbf{x}_j = [x_{1j}, x_{2j}, \dots, x_{Kj}]^T$ by the SCMA encoder according to a constellation $\mathcal{X}_j$ of size M . In particular, each dimension of $\mathbf{x}_j$ corresponds to a specific resource. Due to the sparsity of the SCMA system, most dimensions of $\mathbf{x}_j$ are equal to zero. Finally, the symbols of all users are added together and transmitted through the channel:

$$\mathbf{y} = \sum_{j=1}^J \text{diag}(\mathbf{h}_j) \mathbf{x}_j + \mathbf{n}, \quad (2)$$

where $\mathbf{y} = [y_1, y_2, \dots, y_K]^T$ represents the received signal vector, $\mathbf{h}_j = [h_{1j}, h_{2j}, \dots, h_{Kj}]^T$ represents the channel vector for user j , $\mathbf{n} = [n_1, n_2, \dots, n_K]^T$ represents the channel noise vector, and n_k represents the Gaussian noise of the resource k with zero mean and variance σ^2 . After receiving $\mathbf{y}$, an IDD receiver is deployed to obtain decoded bits.

B. Traditional IDD Receiver

As shown in Eq. (2), inter-user interference is inherent in the SCMA system, where user 1 suffers interference from users 2 to 5. Therefore, an IDD receiver is required to mitigate this effect [8]. A traditional IDD receiver for an SCMA system consists of an SCMA detector and an LDPC decoder implemented by the belief propagation (BP) algorithm. In this paper, an MPA-based SCMA detector is considered to achieve near-optimal performance. The convergence of MPA has been thoroughly discussed in [8] and [23]. To avoid confusion, we define the outer iteration as the iteration between the detector and the decoder, while the MPA iteration and BP iteration refer to the inner iterations within the detector and decoder, respectively.

The MPA-based detector performs soft demodulation by iteratively exchanging information between UNs and RNs. Upon receiving $\mathbf{y}$, the detector first initializes $\text{Pr}_{a,j}(\mathbf{x}_j)$, which is the log-domain *a-priori* probability of symbol $\mathbf{x}_j$ of the j -user. In the first outer iteration, the probabilities of each $\mathbf{x}_j$ are considered to be identical, given as [8]

$$\text{Pr}_{a,j}(\mathbf{x}_j) = \log \left(\frac{1}{M} \right), \quad \forall j = 1, \dots, J, \quad (3)$$

In the following outer iteration, $\text{Pr}_{a,j}(\mathbf{x}_j)$ is updated based on the *a-posteriori* log-likelihood ratios (LLRs) $L_{p,j}$ provided by the LDPC decoder, and its equation is explicitly given in Eq. (10). The log-domain information passed from the j -th UN to the k -th RN in the t -th MPA iteration is defined as $Q_{j \rightarrow k}^{\text{det},t}(\mathbf{x}_j)$. Before the MPA process begins, this message is initialized as [8]

$$Q_{j \rightarrow k}^{\text{det},0}(\mathbf{x}_j) = \text{Pr}_{a,j}(\mathbf{x}_j). \quad (4)$$

Subsequently, the MPA iteration is carried out. Each t -th MPA iteration consists of three steps as follows.

1) Step 1: Passing information from RNs to UNs

The log-domain information passed from the k -th RN to the j -th UN, denoted by $L_{k \rightarrow j}^{\text{det},t}(\mathbf{x}_j)$, is computed as [24]

$$L_{k \rightarrow j}^{\text{det},t}(\mathbf{x}_j) = \log \sum_{\mathbf{x}_i} \exp \left[-\frac{1}{\sigma^2} \|y_k - \sum_{x_{kj} \neq 0} h_{kj} x_{kj}\|^2 \right] + \sum_{i \in \mathcal{N}(k) \setminus j} Q_{i \rightarrow k}^{\text{det},t-1}(\mathbf{x}_i), \quad (5)$$

where $\mathcal{N}(k) \setminus j$ represents the set of users transmitted on resource k except user j .

2) Step 2: Passing information from UNs to RNs

The log-domain information passed from the j -th UN to the k -th RN $Q_{j \rightarrow k}^{\text{det},t}$ is computed as [24]

$$Q_{j \rightarrow k}^{\text{det},t}(\mathbf{x}_j) = \text{Pr}_{a,j}(\mathbf{x}_j) + \sum_{i \in \mathcal{R}(j) \setminus k} L_{k \rightarrow j}^{\text{det},t}(\mathbf{x}_j), \quad (6)$$

where $\mathcal{R}(j) \setminus k$ represents the set of resources carrying user j except resource k .

3) Step 3: Calculating the symbol LLRs

The normalization is performed to ensure that the sum of probabilities of all symbols is equal to one [24]

$$Q_{j \rightarrow k}^{\text{det},t}(\mathbf{x}_j) = Q_{j \rightarrow k}^{\text{det},t}(\mathbf{x}_j) - \log \sum_{\mathbf{x}_j} \exp(Q_{j \rightarrow k}^{\text{det},t}(\mathbf{x}_j)). \quad (7)$$

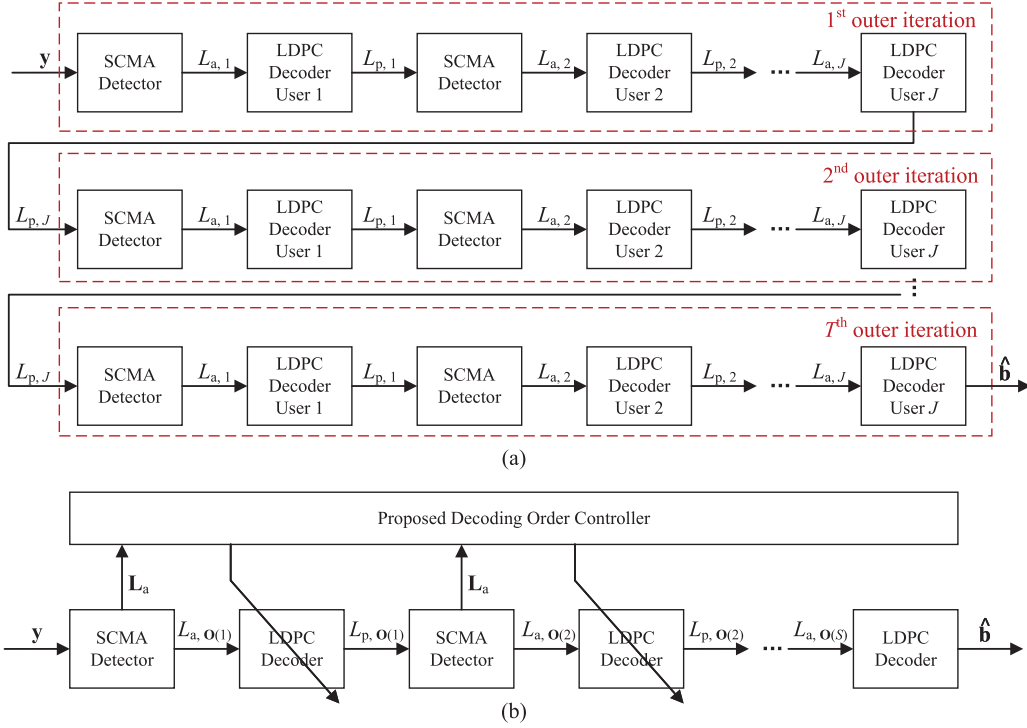


Fig. 3. Block diagram of (a) the proposed SDD receiver and (b) the proposed R-SDD receiver.

The log-domain *a-posteriori* probabilities of each symbol $\mathbf{x}_j$, denoted by $\Pr_{p,j}^t(\mathbf{x}_j)$, can be computed as

$$\Pr_{p,j}^t(\mathbf{x}_j) = \Pr_{a,j}(\mathbf{x}_j) + \sum_{k \in \mathcal{R}(j)} L_{k \rightarrow j}^{\text{det}, T_{\text{MPA}}}(\mathbf{x}_j). \quad (8)$$

The MPA iteration terminates upon reaching the maximum number of MPA iterations T_{MPA} . Finally, $\Pr_{p,j}^t(\mathbf{x}_j)$ is converted to the *a-priori* LLR for the subsequent LDPC decoder [24], which is denoted as $L_{a,j}$:

$$L_{a,j}(c_j^i) = \log \frac{\sum_{\{\mathbf{x}_j \in \mathcal{X} | c_j^i = 0\}} \Pr_{p,j}^t(\mathbf{x}_j)}{\sum_{\{\mathbf{x}_j \in \mathcal{X} | c_j^i = 1\}} \Pr_{p,j}^t(\mathbf{x}_j)}, \quad (9)$$

where c_j^i represents the i -th bit associated with symbol $\mathbf{x}_j$ of the j -th user.

Once the MPA-based detector calculates the *a-priori* LLRs for each user, the LDPC decoder is deployed to obtain the decoded bits and the *a-posteriori* LLRs $L_{p,j}$. The *a-posteriori* LLRs $L_{p,j}$ are then utilized to compute $\Pr_a(\mathbf{x}_j)$ for the detector in the next outer iteration, which is given as:

$$\Pr_{a,j}(\mathbf{x}_j) = \log \left\{ \prod_{i=1}^{\log_2 M} \frac{1}{1 + \exp[(-1)^{c_j^i} \cdot L_{p,j}(c_j^i)]} \right\}. \quad (10)$$

A complete outer iteration consists of one MPA-based detector and J LDPC decoders. The iteration process continues until all users are successfully decoded or the maximum number of outer iterations T_{out} is reached. At last, the final decoded results are output.

The main drawbacks of the traditional IDD receiver can be summarized as follows: Within one outer iteration, a user

cannot benefit from the decoding results of other users. Additionally, successfully decoded users continue to be processed in outer iterations, even though their contributions to inter-user information exchange are minimal. This unnecessary decoding process consumes a significant amount of computational resources, leaving room for further optimization in receiver design.

III. PROPOSED R-SDD RECEIVER AND FG INTERLEAVING SCHEME

The proposed R-SDD receiver follows a two-stage design. In stage one, we propose the successive detection and decoding (SDD) receiver to accelerate convergence and enhance system performance, as detailed in Sect. III-A. Based on this structure, stage two introduces the decoding order controller, which is discussed in Sect. III-B. To further reduce receiver complexity, we employ an early-stopping criterion for the MPA-based detector in Sect. III-C. Finally, we design a FG interleaving scheme, as presented in Sect. III-D.

A. Successive Detection and Decoding Receiver

The block diagram of the SDD receiver is shown in Fig. 3(a). The decoders of different users are operated successively.¹ After receiving the signal y , the receiver initializes

¹An SIC-based receiver has been proposed in [12] tailored for SCMA systems. However, this work focuses on eliminating codebook collisions in grant-free scenarios by performing SIC between the collision-resolution and detection modules. In contrast, our proposed R-SDD receiver mitigates inter-user interference via SDD between the detector and the decoder. Therefore, the proposed R-SDD receiver is fundamentally different from the SIC-based receiver in [12].

Algorithm 1 Proposed R-SDD Receiver

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1 Deploy SCMA detector to obtain  $L_{a,j}$  of all  $J$  users;
2 Calculate  $\text{AMI}_{p,j}^{\text{user}}$  of all  $J$  users by Eq. (11);
3 Sort  $\text{AMI}_{p,j}^{\text{user}}$  and record the order in  $\mathbf{O}$ , where  $\mathbf{O}(s) = j$ 
  denotes user  $j$  has the  $s$ -th largest AMI;
4 for  $s = 1, 2, \dots, J$  do
5   if  $s \neq 1$  then
6     Deploy SCMA detector to obtain  $L_{a,\mathbf{O}(s)}$ ;
7   end
8   Deploy LDPC decoder to obtain  $L_{p,\mathbf{O}(s)}$ ;
9 end
10 Record the users violating the parity-check constraints,
    and let  $S$  be their number;
11 if  $S=0$  then
12   Output decoding results;
13 else
14   for  $t = 2, 3, \dots, T_{\text{out}}$  do
15     Update  $\text{AMI}_{p,j}^{\text{user}}$  of these  $S$  users and sort them in
      descending order to update  $\mathbf{O}$ ;
16     for  $s = 1, 2, \dots, S$  do
17       Deploy SCMA detector to obtain  $L_{a,\mathbf{O}(s)}$ ;
18       Deploy LDPC decoder to obtain  $L_{p,\mathbf{O}(s)}$ ;
19       Update  $S$ ;
20       if  $S=0$  then
21         Terminate outer iteration;
22       end
23     end
24   end
25   Output decoding results;
26 end

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the extrinsic LLRs to zero, and then sequentially detects and decodes users from 1 to J . Unless the parity-check constraints of all users are fully satisfied, an outer iteration is performed, which repeats the detection and decoding process for users 1 to J until correct codewords of all users are obtained.

Within each outer iteration, the SCMA detector and LDPC decoder operate in pairs. For the j -th user, the MPA-based SCMA detector updates $L_{a,j}$ for the LDPC decoder. The LDPC decoder then performs BP decoding to obtain $L_{p,j}$, and updates the extrinsic LLR for the next SCMA detector. To facilitate the exposition of proposed SDD mechanism, we define that each SDD decoding process consists of an MPA-based detection process and an LDPC decoding process of one user. Compared with the conventional IDD receiver [8], [13], the SDD receiver offers the following advantages:

- 1) Within one outer iteration, subsequent users can benefit from the decoding results of users that have already been decoded, thereby accelerating system convergence. The traditional IDD receiver, however, cannot leverage this information.
- 2) The system can perform an arbitrary number of SDD decoding process, terminating once the slowest-converging user is successfully decoded. This reduces complexity and improves flexibility, as the required number of SDD decoding process does not have to be a multiple of J . On the contrary, the traditional

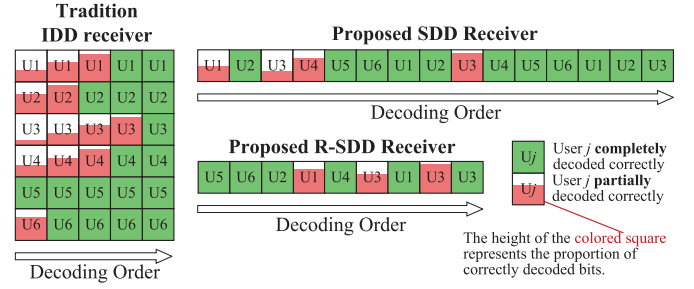


Fig. 4. Example of the decoding orders in traditional IDD receiver, the proposed SDD receiver, and the proposed R-SDD receiver. Assume that user 5 and user 6 have the most reliable information, and the signal of user 3 is heavily interfered.

IDD receiver must always decode all J users per outer iteration.

B. Decoding Order Re-Ranking

As discussed in the previous section, subsequent users benefit from the extrinsic information of already-decoded users, and thus achieve a higher outer-iteration gain in the SDD receiver. Since the performance bottleneck of the SCMA system is always determined by the worst user, placing the worst user at the end of decoding process in each outer iteration can maximize the iterative gain, thereby improving the overall system performance. In this manner, additional computational resources can be allocated to the worst user. Motivated by this idea, we propose a decoding order controller to re-rank the users by their reliability and dynamically adjust the decoding order.

The block diagram of the ranked-SDD (R-SDD) receiver is shown in Fig. 3(b). Unlike the linear decoding order in the standard SDD structure, the decoding order in R-SDD is determined by the decoding order controller, which operates as follows:

- 1) In the first outer iteration, the average mutual information (AMI) of each user is calculated and sorted in descending order. The SDD decoding process starts from the user having the highest AMI.
- 2) In the subsequent outer iteration, only users that have not yet been successfully decoded are considered. The SDD decoding processes are carried out for these users in descending order of their corresponding AMI values. Successfully decoded users preserve their LLR values from the previous outer iteration.

To clearly illustrate the details of the R-SDD receiver, we provide the corresponding pseudocode, as shown in *Algorithm 1*. The AMI between the encoded bits and their corresponding *a-posteriori* LLRs, denoted as $\text{AMI}_{p,j}^{\text{user}}$, is used to evaluate the reliability of each user,² which is given by [25]

$$\text{AMI}_{p,j}^{\text{user}} = \mathbb{E}[f_1(L_{p,j})], \quad (11)$$

²Since the LDPC decoder is not yet deployed during the initial AMI calculation, L_a is used in place of L_p in Eq. (11).

where $\mathbb{E}(s)$ represents the mathematical expectation of the variable s , and $f_1(x)$ is given by:

$$f_1(x) = \frac{1}{1+e^x} \log_2 \frac{2}{1+e^x} + \frac{1}{1+e^{-x}} \log_2 \frac{2}{1+e^{-x}}. \quad (12)$$

As a matter of fact, the value of $\text{AMI}_{p,j}$ depends solely on the absolute values of the *a-posteriori* LLRs and is independent of the actual values of the encoded bits. Therefore, the receiver can compute $\text{AMI}_{p,j}^{\text{user}}$ of each user without knowing the transmitted encoded bits.

After each SDD decoding process, the corresponding $\text{AMI}_{p,j}^{\text{user}}$ must be updated. This strategy dynamically re-ranks the decoding order and improves the overall performance of the SCMA system. Furthermore, computational complexity is also reduced by this approach. In the traditional IDD receiver, even if only one user remains with unsatisfied parity-check constraints, all J users must be decoded in the next outer iteration. In contrast, the R-SDD receiver only need to perform one SDD decoding process for the worst user in the next outer iteration.

To better illustrate the differences between traditional receiver and our proposed receivers, we present an example in Fig. 4. Each square in the figure represents an LDPC decoding process, where green squares indicate the decoded bits satisfy parity-check constraints, while red squares represent unsuccessful SDD decoding processes. The SDD receiver improves the efficiency of inter-user information exchange. The R-SDD receiver further improves performance by re-ranking decoding order to maximize outer-iteration gain. It should also be noted that the R-SDD receiver has the lowest computational complexity.

C. Early-Stopping Criterion for MPA-Based Detector

Early-stopping criterion has been introduced to the MPA-based detector in [26]. However, such criterion relies on the conversion from log-domain to probability-domain, which introduces additional computational complexity. In this paper, we develop a new early-stopping criterion method based on the variation of LLRs to terminate iterations with relatively lower complexity. Fig. 5 provides an example of LLR variation in the MPA-based detector across different iterations. It can be observed that in the first outer iteration, the first MPA iterations contribute the most significant LLR variation. This indicates that a large portion of MPA iterations are inefficient. Nevertheless, simply restricting the number of MPA iterations may lead to performance degradation, since the required number of MPA iterations can vary across different SCMA symbols and different outer iterations.

To resolve this problem, we propose an early-stopping criterion based on LLR variation to terminate MPA iteration when information exchange is sufficient. We define the average LLR variation within the t -th MPA iterations as V_{avg}^t , given by

$$V_{\text{avg}}^t = \frac{\sum_j \sum_{\mathbf{x}_j} \|\text{Pr}_{p,j}^t(\mathbf{x}_j) - \text{Pr}_{p,j}^{t-1}(\mathbf{x}_j)\|}{J \times M}, \quad (13)$$

where $\text{Pr}_{p,j}^0(\mathbf{x}_j)$ is considered to be 0 before the MPA iteration.

If the V_{avg}^t of the current MPA iteration falls below a predefined threshold τ_{MPA} , we assume that the information exchange

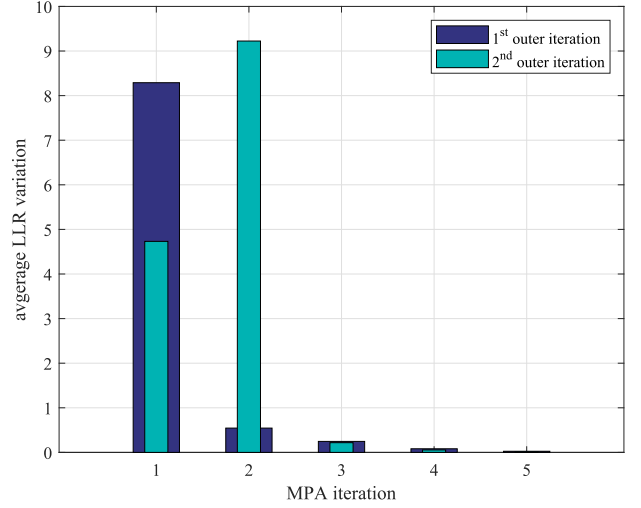


Fig. 5. Average absolute LLR variation in MPA-based detector across different MPA iterations over the AWGN channel, where the traditional IDD receiver is considered and E_b/N_0 is set to 3.6 dB.

between RNs and UNs has reached sufficient convergence, and the MPA iteration is terminated in advance. Actually, we have conducted various experiments to verify that the computational complexity of the MPA-based detector can be effectively reduced without compromising performance when τ_{MPA} changes from 0 to 1. Nonetheless, when $\tau_{\text{MPA}} > 1$, the system performance will deteriorate significantly. In consequence, we set $\tau_{\text{MPA}} = 1$ in the proposed early-stopping criterion.

D. FG Interleaving Scheme

In coding theory, increasing the irregularity of LDPC codes can enhance their waterfall performance [27]. Consequently, in practical communication systems, such as 5G and WiFi, irregular LDPC codes are deployed to achieve optimal performance [28], [29]. Mapping high-performance variable nodes (VNs) to low-performance labeling bits can provide unequal protection and enhance performance in BICM systems [30]. Nonetheless, in most SCMA codebooks, there is no significant reliability difference among the labeling bits. The traditional SCMA interleaver [8], which applies the same mapping design to all users, limits the overall performance of SCMA systems. To address this issue, we design an FG interleaving scheme based on the structure of the factor graph, specifically tailored for SCMA systems. The novelty of the proposed FG interleaving scheme lies in two main aspects: using mutual information to more accurately evaluate the reliability of VNs, and adopting a mapping strategy based on the FG structure.

The traditional variable-node-degree-based (VND) method, which assumes that VNs with higher degrees are more reliable, has been widely adopted in interleaving designs for coded modulation systems [31], [32]. However, the existing VND method has the following main drawbacks:

- 1) The VND method cannot distinguish between different variable nodes with the same degree.
- 2) The VND method determines reliability based on the degree of VN, which may be inaccurate as it does

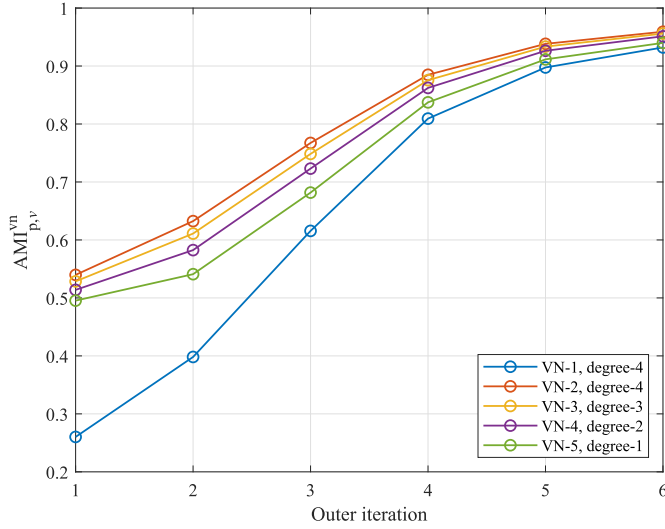


Fig. 6. AMI of different VNs in IARA protograph code across outer iterations in the SCMA systems with traditional IDD receiver, where AWGN channel is considered and E_b/N_0 is set to 2.6 dB.

not consider specific communication system. In certain iterative systems, the VN with the highest degree does not always converge the fastest [33].

To overcome these limitations, we use the AMI of each VN to evaluate their convergence behaviors. Precisely speaking, the VNs with higher AMI possess higher reliability. Let $\text{AMI}_{p,v}^{\text{VN}}$ denote the average *a-posteriori* mutual information of VN- v . It is computed offline from the simulated LLRs $L_{p,j}^v$ using the Monte Carlo method. Concretely, we simulate a sufficiently large number of $L_{p,j}^v$ samples to estimate the mean of $f_1(L_{p,j}^v)$. Consequently, $\text{AMI}_{p,v}^{\text{VN}}$ can be calculated as

$$\text{AMI}_{p,v}^{\text{VN}} = \mathbb{E}[f_1(L_{p,j}^v)], \quad (14)$$

where $L_{p,j}^v$ represents the *a-posteriori* LLRs of VN- v corresponding to the j -th user, and VN- v represents the v -th VN of the corresponding protograph. To ensure the accuracy of AMI analysis, the SNR should be chosen within the waterfall region.

Fig. 6 illustrates the AMI convergence behavior of different VNs across iterations for the improved accumulate-repeat-accumulate (IARA) protograph code [34]. It can be observed that although VN-1 and VN-2 both have a degree of 4, their convergence speeds are different. In particular, VN-1 has the slowest convergence speed. On the contrary, the conventional VND method treats these two VNs as having identical convergence properties, disregarding their differences. By leveraging AMI to evaluate the convergence speed of different VNs, we can not only distinguish VNs with the same degree, but also ensure that the assessment of user reliability is consistent with the modulation schemes.

The goal of the FG interleaver is to ensure that each SCMA symbol is mapped to VNs with different AMI values. In this way, during the outer iterations, the VNs with lower reliability can benefit from highly reliable extrinsic information associated with other VNs provided by the detector, thereby accelerating the convergence of the SCMA system. Specifically, the J users are partitioned into G groups according

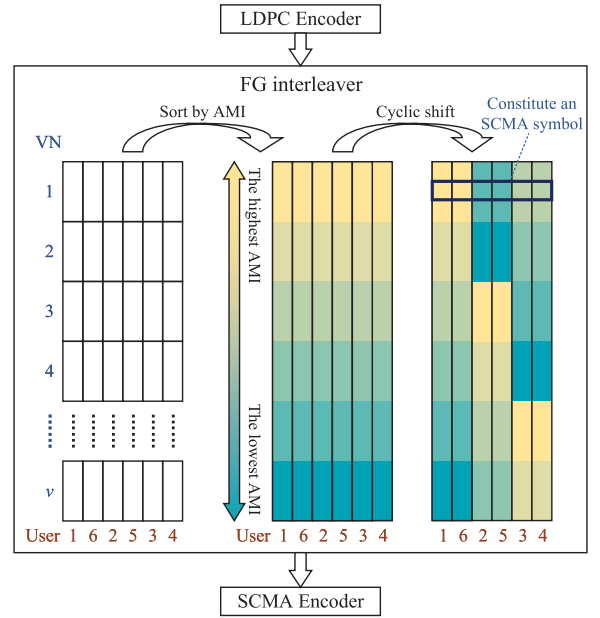


Fig. 7. FG interleaving scheme in 4×6 SCMA system.

to the factor graph, where each group contains c mutually orthogonal users and thus $G = J/c$. Since one user within the same group are orthogonal and cannot exploit the extrinsic information of other users during detection, they are allowed to share the same mapping. For each LDPC codeword in group g ($g = 1, 2, \dots, G$), the FG interleaver first sorts the VNs in descending order of AMI, and then applies a cyclic shift of length $g \cdot n/G$. This shift guarantees that different groups adopt different mappings, ensuring that each SCMA symbol is composed of VNs with diverse reliabilities. As a result, each VN is connected to other VNs with different reliability levels through the detector. An illustrative example of the proposed FG interleaver for a 4×6 SCMA system is provided in Fig. 7. In this case, six users are divided into three groups, and each modulation symbol is mapped to VNs with low, medium, and high AMI, respectively. By interleaving VNs, we can maximize the efficiency of information exchange and accelerate the convergence of the symbols containing the weakest VNs.

IV. MULTI-USER PROTOGRAPH CODE DESIGN FOR SCMA SYSTEMS

Due to the inherently overloaded nature of SCMA, the receiver heavily relies on iterative receiver to achieve desirable performance. Furthermore, the SCMA receiver requires the well-designed code to provide sufficient iteration gain. However, conventional code design usually focuses on convergence behavior without considering outer-iteration gain. This motivates us to design codes specifically tailored for the multi-user scenario.

A. Background

Recent studies on LDPC code design for NOMA systems primarily fall into two categories. The first category focuses on irregular LDPC codes based on degree distribution. Due

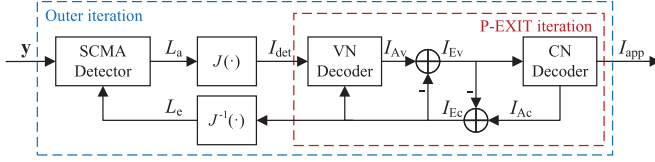


Fig. 8. Block diagram of modified P-EXIT algorithm.

to the high complexity of analyzing multi-user scenarios, these works typically consider only two users [20], [21], [35], which violates the high overloading factor that SCMA aims to achieve. Another category focuses on designing LDPC codes based on protograph. However, these works focus on other scenarios such as NOMA and GMAC [19], [22], [36], rather than SCMA.

The main drawback of existing code designs is that they focus solely on decoder performance, while neglecting the gain from outer iterations. By contrast, for overloaded multi-user systems such as SCMA, iterative receivers are essential to eliminate inter-user interference. As a result, outer-iteration gain plays a critical role in overall performance. Therefore, existing LDPC codes are not suitable for such systems. To address this limitation, we investigate the design of protograph LDPC codes tailored for SCMA systems.

B. Modified P-EXIT Algorithm for SCMA System

The P-EXIT algorithm, which was first introduced in [37], has become a powerful tool for analyzing the convergence behavior of protograph codes in a specific coded-modulation system. As shown in Fig. 8, by analyzing the mutual information (MI) exchange between the VN decoder and CN decoder, one can determine the decoding threshold without requiring complex simulations. The main difference between our design and the conventional P-EXIT algorithm is that we trace the MI updates during the outer iterations in the SCMA receiver, but not only during the BP iterations. This allows us to not only evaluate the error performance of the code under the BP decoding, but also evaluate the outer-iteration gain, which is important for SCMA receivers.

For a clear description, we define the following MIs:

- I_{Av} denotes the *a-priori* MI between the *a-priori* LLR L_a and the corresponding coded bit.
- I_{Ec} denotes the extrinsic MI between L_{Ec} and the corresponding coded bit, where L_{Ec} represents the output extrinsic LLR sent by CN decoder to VN decoder.
- I_{app} denotes the *a-posteriori* MI between the *a-posteriori* LLR L_p and the corresponding coded bit.

The detailed process of the modified P-EXIT algorithm is shown as follows:

- 1) *Initialization*: Set maximum number of outer iterations T_{out} and number of P-EXIT inner iteration T_{P-EXIT} . In this paper, we consider $T_{out} = 4$ and $T_{P-EXIT} = 100$.
- 2) *Calculating extrinsic LLR L_e* : For the first outer iteration, L_e is initialized to 0. For the subsequent outer iteration, assume that L_e follows a Gaussian distribution $L_e \sim \mathcal{N}(\mu_e, \sigma_e^2)$, where $\mu_e = \frac{\sigma_e^2}{2}$ when the coded bit is 0, and $\mu_e = -\frac{\sigma_e^2}{2}$ when the coded bit is 1. The variance

σ_e of L_e can be calculated as $\sigma_e = J^{-1}(I_{Ec})$. Here, the $J(\cdot)$ function and its inverse function $J^{-1}(\cdot)$ are defined in [18], and $J(\cdot)$ function is given by

$$J(\sigma) = 1 - \int_{-\infty}^{\infty} \frac{\exp\left(-\frac{(\mu - \sigma^2/2)^2}{2\sigma^2}\right)}{\sqrt{2\pi\sigma^2}} d\mu. \quad (15)$$

Once σ_e is obtained, we use it to compute μ_e and determine the value of $L_e = \mu_e$.

- 3) *Calculating the a-priori MI I_{Av}* : First, the SCMA detector is employed to calculate L_a based on y and L_e as inputs. Second, calculate the variance of L_a , which is denoted as σ_a . Third, the mutual information I_{Av} is calculated as $I_{Av} = J(\sigma_a)$. For punctured VNs, the corresponding I_{Av} are set to 0.
- 4) *Updating MIs between VN decoder and CN decoder*: Since we do not modify the MIs exchange process between VN decoder and CN decoder, we refer readers to [18] for more details. The MIs are calculated iteratively until the number of P-EXIT iterations reaches T_{P-EXIT} .
- 5) *Calculating the a-posteriori MI I_{app}* : If I_{app} of all VNs reach 1, this indicates that the LDPC code can converge at the current signal noise ratio (SNR), and the outer iteration then terminates. Otherwise, go back to step 2).

The minimum SNR that allows the protograph-coded SCMA system to converge is defined as the decoding threshold, which can be utilized to evaluate the performance of a protograph code in the waterfall region.

C. Design Principle for Multi-User Protograph Code

In the conventional protograph design, the following approaches are commonly used to enhance performance: Using precoding structures, puncturing high-degree VNs, and controlling the number of degree-2 VNs within a certain range to lower the decoding threshold without introducing an error floor [18]. However, such rules no longer suit for overloaded multi-user scenarios. Through our analysis, we find that a well-designed protograph code for SCMA systems exhibits several new properties. We summarize the unique design principle of the multi-user protograph code in SCMA systems as follows:

1) Avoiding precoding structure

The precoding structure consists of a CN connected to one degree-1 VN and one high-degree VN, where the high-degree VN is punctured. In traditional coding theory, the precoding structure is useful to reduce the decoding threshold of a given protograph [38]. However, such approach significantly deteriorates the outer-iteration gain. This drawback limits the overall performance of these protograph codes in the communication systems employing iterative receivers.

To illustrate this phenomenon, we use the repeat-jagged-accumulate (RJA) code and the accumulate-repeat-by-4-jagged-accumulate (AR4JA) code as examples. As shown in Fig. 9, the RJA code can be regarded as AR4JA code with its precoding structure removed. Fig. 10 compares their convergence behaviors in terms of I_{app} . The dashed lines show I_{app}

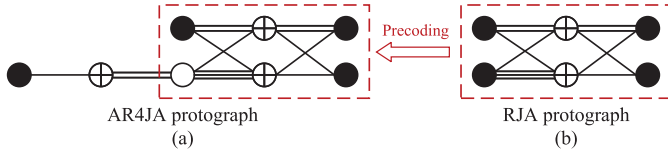


Fig. 9. The protographs of (a) AR4JA code and (b) RJA code, as examples of precoding structure.

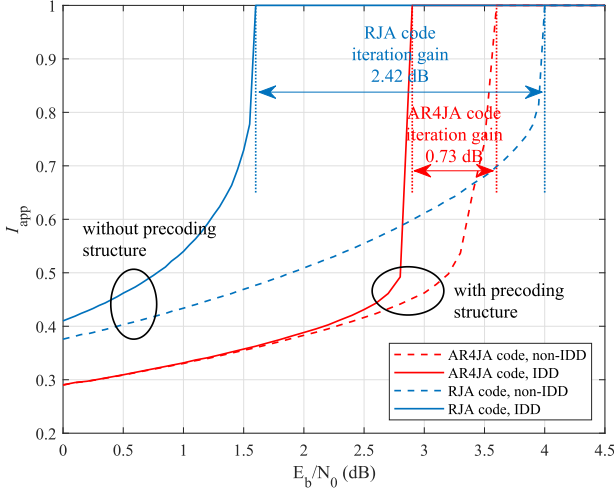


Fig. 10. The *a posteriori* MI convergence trajectories of the rate-1/2 AR4JA code and RJA code over an AWGN channel with and without the iterative receiver, where $T_{\text{out}} = 4$ in the IDD.

without outer iterations, while the solid lines correspond to I_{app} with $T_{\text{out}} = 4$. The gap between these two curves represents the outer-iteration gain. The conventional P-EXIT algorithm is only applicable to evaluate the performance during the initial outer iteration, which coincides with the behavior shown by the dashed lines. However, the AR4JA code only achieves an outer-iteration gain of 0.73 dB, while the RJA code achieves a significantly higher outer-iteration gain of 2.42 dB. This makes the RJA code more compatible with SCMA.

We have further investigated other protograph codes that include precoding structure, such as the IARA code [34] and accumulate-repeat-by-3-accumulate (AR3A) code [39] and observed similar phenomenon. In fact, the outer iterations between the detector and decoder is essential for mitigating inter-user interference in SCMA systems. Punctured VNs, which are not transmitted over the channel, cannot benefit from the mutual information update during outer iterations, and thus reduce the achievable outer-iteration gain. As a matter of fact, using a precoding structure to improve the decoding threshold has become a popular practice in modern protograph design [30], [40], [41]. Nonetheless, we demonstrate that it is essential to avoid the precoding structure in order to design an effective protograph for inter-user interference scenarios.

2) Increasing the proportion of degree-2 VNs

In traditional code design, it is widely recognized that increasing the proportion of degree-2 VNs can lower the decoding threshold and improve waterfall performance. However, too many degree-2 VNs limit the linear growth of the asymptotic minimum distance with code length, resulting in an error floor [38]. For example, both the AR3A code and

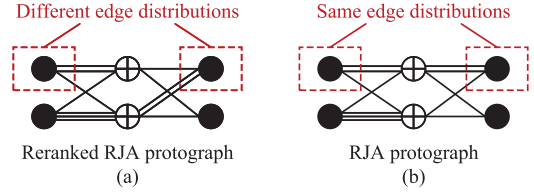


Fig. 11. The protographs of (a) reranked RJA code and (b) RJA code, to illustrate different edge distributions.

the accumulate-repeat-accumulate (ARA) code include 40% degree-2 VNs. Although they can achieve very low decoding thresholds, their BER curves still converge slowly at high SNR region. To address this issue, existing code designs typically limit the proportion of degree-2 VNs to 30% or less. This constraint restricts the design of capacity-approaching protograph codes.

However, in the iterative receiver, the system performance is not only determined by one single BP decoding process. In particular, for multi-user communication systems, each symbol carries coded bits from multiple users. These coded bits associated with different VNs and can exchange information through outer iterations. As a result, degree-2 VNs can receive extrinsic information from higher-degree VNs in other codewords. This allows the trapping set problem in single decoding process to be mitigated by outer iterations. Consequently, the error floor problem can be substantially alleviated in the iterative receiver of multi-user systems. Therefore, multi-user codes can employ a higher proportion of degree-2 VNs to improve performance and further reduce the decoding threshold. In 4×6 SCMA systems, even a 50% proportion of degree-2 VNs does not lead to an error floor.

3) Assigning different VNs with distinct edge distributions

Irregular edge distributions can further enhance the outer-iteration gain of a protograph. During outer iterations, the detector provides extrinsic information with varying statistical characteristics for decoder. In a protograph code, if two VNs share the same edge distribution, they connect to CNs in exactly the same way. These two VNs will update their corresponding extrinsic information via identical paths in the decoder, leading to highly similar reliability. Consequently, one VN cannot converge faster to assist the other through BP iteration. Using distinct edge distributions can better utilize the reliability difference of the extrinsic information output from the detector, so as to accelerate the convergence of the decoder. However, in the non-iterative receiver, the extrinsic information is passed from the detector to the decoder only once, and thus the reliability difference among different VNs is relatively small. For this reason, the distinct edge distributions are not effective to improve the performance of traditional code design.

We use the RJA protograph and a reranked RJA protograph as an example, whose structures are shown in Fig. 11. Both protographs contain two degree-3 VNs, and each of them connects to two CNs through one degree-1 edge and one degree-2 edge. The difference between these protographs lies in edge distributions: in the reranked RJA protograph, the degree-2 edges connect to different CNs, while in the original

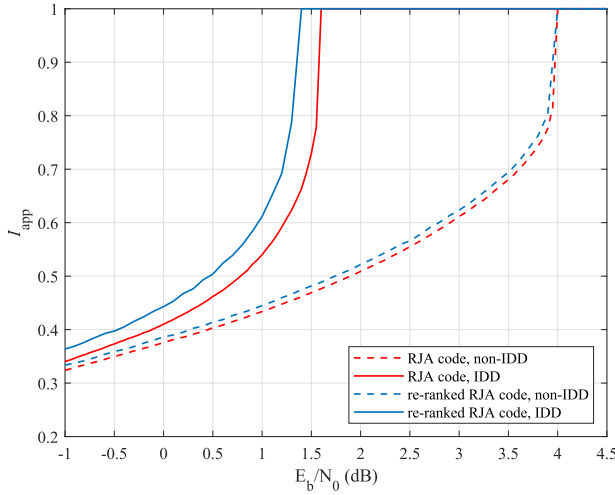


Fig. 12. The *a posteriori* MI convergence trajectories of the rate-1/2 RJA code and reranked RJA code over an AWGN channel with and without the iterative receiver, where $T_{\text{out}} = 4$ in the IDD.

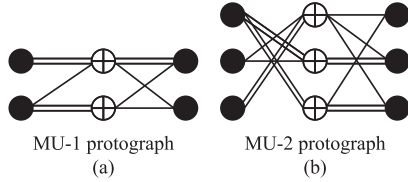


Fig. 13. The proposed protographs of (a) MU-1 code and (b) MU-2 code.

RJA protograph, these two degree-2 VNs share the same edge distribution.

Fig. 12 compares the convergence performance of these two codes in terms of I_{APP} , with and without an iterative receiver. As shown, the edge distribution has minimal impact on the initial decoding process. Therefore, in the traditional single-user code design, edge distribution does not attract much attention. On the contrary, edge distribution has significant impact on the outer-iteration gain. The reranked RJA protograph yields approximately 0.2 dB improvement over the original RJA code. This highlights the importance of using distinct edge distributions when designing multi-user codes.

Remark 1: As proved in [42], LDPC codes exhibit similar performance over AWGN and other memoryless channels. Hence, although an AWGN channel is assumed in our design, the proposed principle is also applicable to ergodic channels such as fast fading channels.

D. Proposed Protograph Codes

Following the design principle mentioned above, we propose two rate-1/2 multi-user (MU) protograph codes, i.e., called MU-1 and MU-2 codes, as shown in Fig. 13. These codes have no precoding structure and keep distinct edge distributions for different VNs. Additionally, these codes have a 50% proportion of degree-2 VNs. To limit the decoding complexity, we set the maximum degree of the VNs to 4. P-EXIT algorithm is deployed to search for good candidates under the above constraints. Finally, an optimal 2×4 pro-

TABLE I
DECODING THRESHOLDS OF DIFFERENT PROTOGRAPH CODES IN 4×6 SCMA SYSTEM OVER THE AWGN CHANNEL

Protograph code	Threshold $T_{\text{out}} = 1$	Threshold $T_{\text{out}} = 4$	Iteration gain
Proposed MU-1 code	3.92 dB	1.14 dB	2.78 dB
Proposed MU-2 code	4.17 dB	0.98 dB	3.19 dB
GMAC code [22]	3.50 dB	1.28 dB	2.22 dB
RJA code [33]	4.02 dB	1.60 dB	2.42 dB
Regular-(3, 6) code [18]	4.17 dB	1.74 dB	2.43 dB
IARA code [34]	3.51 dB	1.78 dB	1.73 dB
AR4JA code [39]	3.65 dB	2.92 dB	0.73 dB

toph, called MU-1 protograph, is obtained by exhaustive search. The corresponding base matrix is given by

$$\mathbf{B}_{\text{MU-1}} = \begin{bmatrix} 2 & 1 & 2 & 1 \\ 0 & 2 & 1 & 1 \end{bmatrix}. \quad (16)$$

Based on MU-1, we extend the design to a 3×6 protograph, i.e., MU-2 protograph. Due to a larger search space and greater irregularity, the error correction capability of MU-2 protograph is better than MU-1 protograph. The corresponding base matrix of MU-2 protograph is given as follows:

$$\mathbf{B}_{\text{MU-2}} = \begin{bmatrix} 0 & 1 & 2 & 1 & 1 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 0 & 1 & 0 & 2 \end{bmatrix}. \quad (17)$$

To evaluate the superiority of the proposed protograph codes compared with the existing designs, we compute their decoding thresholds using the modified P-EXIT algorithm, as shown in Table I. Among the listed protograph codes, the proposed MU-1 and MU-2 codes demonstrate a superior performance. As typical competitors, the AR4JA and IARA codes suffer severely from the precoding structure. Specifically, the AR4JA and IARA codes obtain less than 2 dB outer-iteration gain with $T_{\text{out}} = 4$, causing poor overall convergence performance. Meanwhile, around 3 dB outer-iteration gain can be obtained by our proposed MU-1 and MU-2 codes. The RJA and regular-(3, 6) codes with higher iteration gain are limited by the proportion of degree-2 VNs, and thus cannot obtain ideal overall performance either. Compared to existing protographs, the proposed MU-1 code achieves at least a 0.3 dB gain. The proposed MU-2 protograph code achieves further superior performance with respect to the MU-1 code. It is worth noting that although the GMAC code in [22] exhibits an excellent decoding threshold, it suffers from a severe error floor in SCMA systems.

V. COMPLEXITY ANALYSIS

In this section, we analyze the computational complexity of the proposed R-SDD receiver. In protograph-coded SCMA systems, the primary computational complexity comes from the SCMA detector and the BP decoder. The proposed R-SDD structure gives priority to decoding more reliable users. This strategy improves the efficiency of extrinsic information exchange and reduces the number of outer iterations. Moreover, the R-SDD receiver improves the reliability of the *a priori* LLRs, which enables the BP decoder to converge in fewer iterations and thus reduces the computational complexity

of the BP decoder. As in other coded systems, BP iterations contribute the majority of the computational complexity in SCMA receivers. Consequently, the combination of fewer outer iterations and reduced BP iteration allows the proposed R-SDD receiver to achieve lower overall complexity.

To provide a more detailed complexity analysis, we measure the average number of operations required by both the proposed R-SDD receiver and the traditional IDD receiver over an AWGN channel. Both BP and MPA algorithms are implemented in the log domain, converting multiplications into additions. Additionally, the BP decoder requires tanh operation, which is typically implemented using a look-up table in practical systems. We account for tanh operation separately in our analysis. For a fair comparison, we define the maximum decoding attempts as T_{dec} , which is the maximum number of LDPC decoding processes that can be performed in the receiver, accumulated over all outer iterations. Once the number of decoding processes reaches T_{dec} , the outer iterations terminate regardless of whether decoding is successful. In a traditional IDD receiver, $T_{\text{dec}} = T_{\text{out}} \times J$, where T_{out} is the number of outer iterations. In contrast, in the proposed R-SDD receiver, T_{dec} can be arbitrarily specified. Complexity and performance of different receivers are compared under the same T_{dec} .

The computational complexity of different receivers is illustrated in Fig. 14. As discussed earlier, the R-SDD receiver reduces the number of BP iterations, leading to the lowest number of tanh operation. Regarding addition operations, in the SNR region where successful decoding is achievable (i.e., $E_b/N_0 > 2$ dB), the R-SDD receiver exhibits lower complexity compared with the traditional IDD receiver with the same T_{dec} . In the very low SNR region, the addition complexity of R-SDD exceeds that of the traditional IDD receiver. This is because the system has not yet converged at such SNR levels and thus suffers from additional detection costs. However, this SNR region is typically not of concern in most SCMA systems, as reliable communication cannot be achieved under this condition. While the proposed R-SDD receiver benefits from lower complexity with respect to the traditional IDD receiver, the SIC-like structure leads to increased latency, which remains as an important aspect for further investigation.

VI. SIMULATION RESULTS

In this section, we present simulations to validate the proposed scheme. The maximum number of MPA iterations and BP iterations are set to 5 and 100, respectively. To ensure fairness, the early-stopping criterion is applied to both the proposed R-SDD receiver and the traditional IDD receiver, with τ_{MPA} set to 1. Unless otherwise specified, the MU-2 protograph code, perfect CSI, and the same codebook as in [43] are assumed in our simulations. All LDPC codes used in this paper have a rate of 1/2 and a codeword length of 2400.

As shown in Fig. 15, a considerable performance gain can be achieved by using the proposed R-SDD scheme over an AWGN channel. Specifically, at a BER of 1×10^{-5} , the R-SDD receiver provides gains of 0.38 dB, 0.55 dB, 0.40 dB and 0.30 dB compared with the traditional IDD receiver

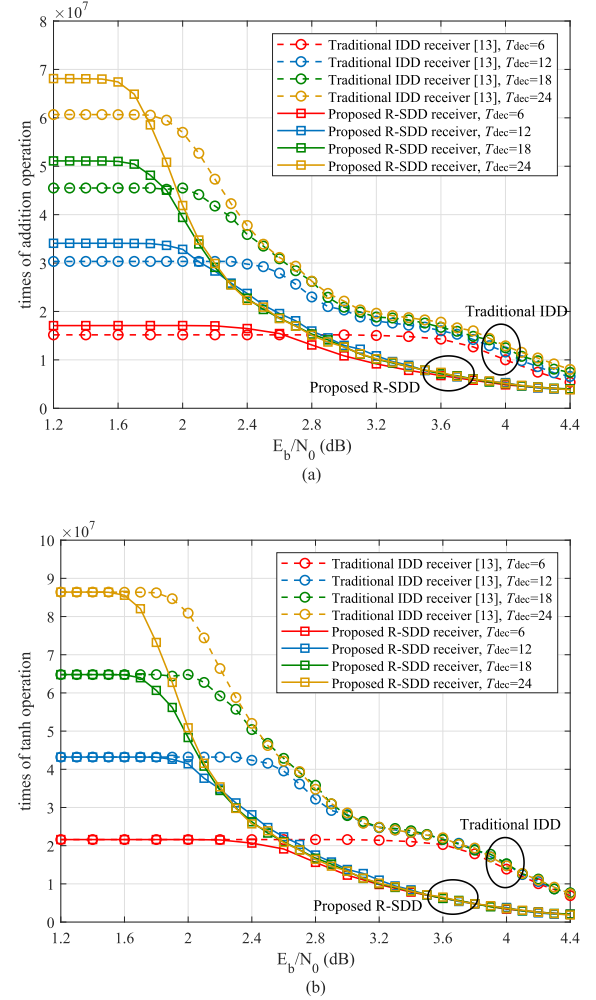


Fig. 14. Computational complexity comparison of 4×6 SCMA systems with the proposed R-SDD receiver and traditional IDD receiver over the AWGN channel, in terms of (a) addition operations and (b) tanh operations within a single codeword.

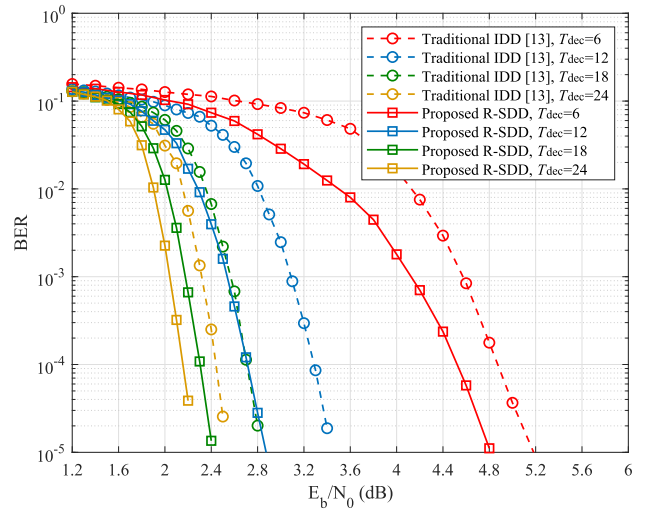


Fig. 15. BER curves of the protograph-coded 4×6 SCMA systems with traditional IDD receiver and proposed R-SDD receiver over an AWGN channel. Different T_{dec} are considered and MU-2 protograph code is adopted.

when $T_{\text{dec}} = 6, 12, 18, 24$, respectively. In most studies, the number of outer iterations is typically set to 4 [8], [13], making

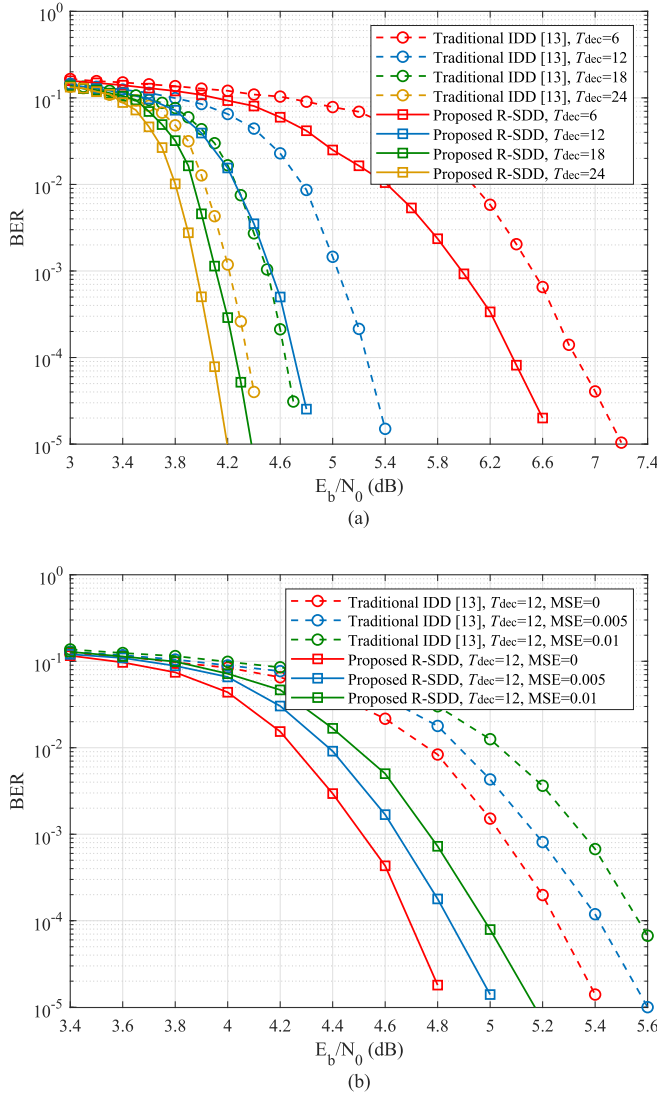


Fig. 16. BER curves of the protograph-coded 4×6 SCMA systems with traditional IDD receiver and proposed R-SDD receiver over (a) Rayleigh fading channel and (b) TDL-B multipath fading channel. Different T_{dec} are considered and MU-2 protograph code is adopted.

$T_{\text{dec}} = 24$ a standard benchmark. Moreover, our proposed R-SDD receiver achieves a desirable performance gain without increasing computational complexity.

We further extend the simulations to more realistic channel models to verify the robustness of the proposed SCMA receiver design in fading scenarios. In particular, we consider both Rayleigh fading and tapped delay line-B (TDL-B) multipath fading channels with the effect of imperfect CSI. For the Rayleigh fading channel, we assume that the fading coefficients are independent across both users and frequencies [8]. For the TDL-B multipath fading channel, we adopt the frequency-selective fading setting in [44]. For a fair comparison, the multipath power is normalized to 1, and mean squared error (MSE) is utilized to depict the CSI estimation error. As shown in Fig. 16(a), over the Rayleigh fading channel, the R-SDD receiver provides gains of 0.40 dB, 0.65 dB, 0.35 dB and 0.20 dB compared with the traditional IDD receiver when $T_{\text{dec}} = 6, 12, 18, 24$, respectively. As can be seen in Fig. 16(b), over the TDL-B multipath fading channel, a consistent gain of

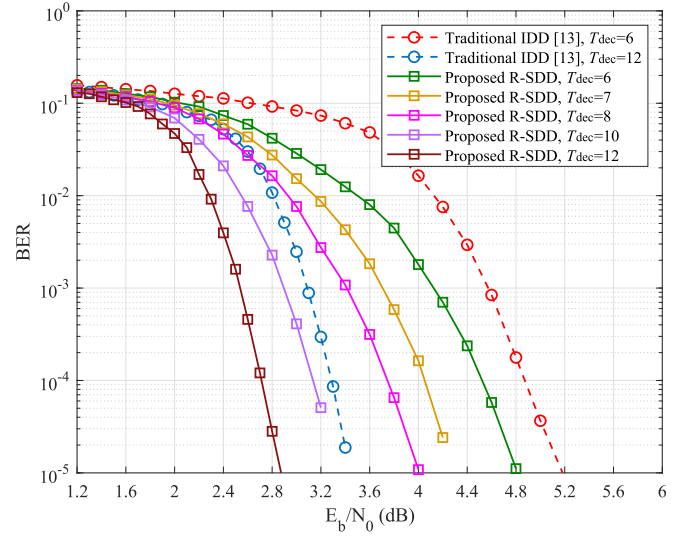


Fig. 17. BER curves of the protograph-coded 4×6 SCMA systems with traditional IDD receiver and proposed R-SDD receiver over an AWGN channel. Different T_{dec} are considered and MU-2 protograph code is adopted.

about 0.5 dB can be obtained when $T_{\text{dec}} = 12$, regardless of CSI accuracy. Compared with the AWGN channel, the gains of the proposed R-SDD receiver under fading scenarios are more remarkable. This is because random fading amplifies the reliability difference among users, thereby enhancing the effect of decoding order re-ranking. These results confirm that the proposed R-SDD receiver can achieve desirable performance advantages in realistic channel scenarios.

To illustrate the flexibility of our design, we compare the BER curves of the proposed R-SDD receiver and the traditional receiver with different T_{dec} in Fig. 17. The R-SDD receiver achieves better BER performance with $T_{\text{dec}} = 10$ than a traditional IDD receiver with $T_{\text{dec}} = 12$. Moreover, in traditional IDD receivers, T_{dec} can only be set as a multiple of the number of users (i.e., J). In contrast, R-SDD allows arbitrary values of T_{dec} . This enables a more flexible trade-off between performance and complexity, making the R-SDD receiver more suitable for hardware implementation.

In fact, practical multicarrier communication systems typically involve a larger number of resources (for example, a 5G resource block contains 12 subcarriers). Thus, we conduct simulations on a larger factor graph and present the result in Fig. 18, where the 12×18 codebook is constructed using the unitary rotation of the QAM constellation as proposed in [45]. One can observe similar phenomenon to the case with 4×6 codebook. In particular, the proposed R-SDD receiver consistently delivers substantial performance gains compared with the traditional IDD receiver.

Fig. 19 presents the BER performance of the protograph-coded SCMA systems with the proposed FG interleaver and traditional interleaver. As can be seen, the proposed FG interleaver provides desirable performance gain compared with the traditional interleaver [8] under different settings. Since the FG interleaver is designed based on the reliability difference of VNs in the protograph, its achievable gain mainly depends on the employed code, but not the receiver and maximum decoding attempts T_{dec} . At a BER of 1×10^{-5} , the proposed

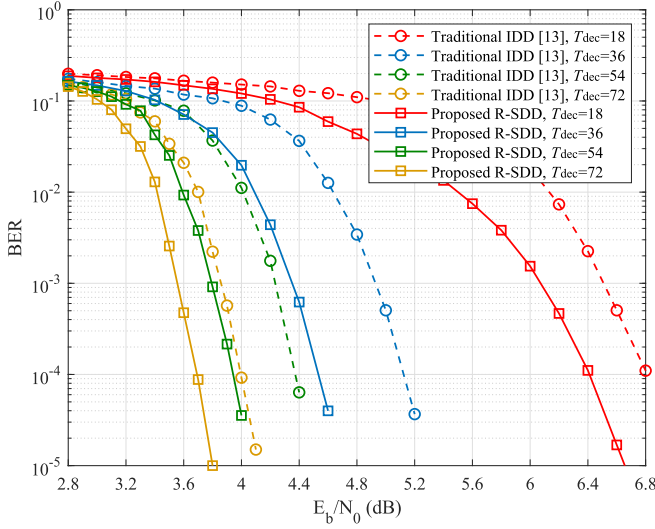


Fig. 18. BER curves of the protograph-coded 12×18 SCMA systems with traditional IDD receiver and proposed R-SDD receiver over the AWGN channel. Different T_{dec} are considered and MU-2 protograph code is adopted.

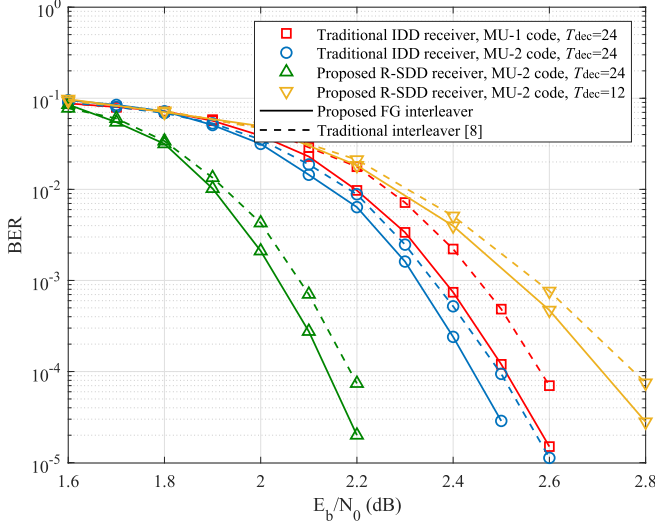


Fig. 19. BER comparison of two different interleaving schemes in protograph-coded 4×6 SCMA systems over the AWGN channel. Different receivers, protograph codes, and T_{dec} are considered.

FG interleaver achieves about 0.07 dB performance gain using the MU-2 protograph code compared with the traditional interleaver. When the MU-1 protograph code is used, the gain increases to about 0.09 dB.

Fig. 20 compares the BER performance of our proposed protograph codes with other state-of-the-art competitors. Since the FG interleaver provides different gains for different code-words, it is not employed here in order to eliminate its influence. The solid lines represent multi-user codes designed for inter-user interference scenarios, while the dashed lines represent the single-user codes optimized for the AWGN channel. It is obvious that single-user codes perform poorly in SCMA systems. This is mainly because they cannot obtain ideal outer-iteration gain, which highlights the importance of code design specifically tailored for SCMA systems.

Among the existing multi-user codes, the GMU code [35] is an irregular LDPC code generated based on degree distribu-

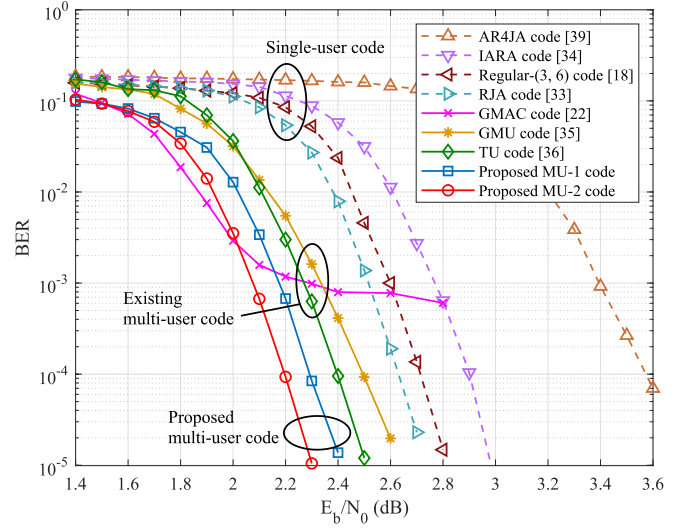


Fig. 20. BER curves of the 4×6 protograph-coded SCMA systems with proposed R-SDD receiver over the AWGN channel, where $T_{\text{dec}} = 24$. Different LDPC codes are considered.

tion, while the others are protograph codes. Compared to these codes, the proposed MU-1 and MU-2 protograph codes exhibit significant performance advantages. In particular, the GMAC code [22] can achieve a acceptable waterfall performance but experiences an early error floor, which limits its applicability in SCMA. This reveals that LDPC codes designed for other NOMA scenarios may not be effective in SCMA. Compared with the TU code [36], the proposed MU-2 protograph code achieves a 0.3 dB gain at a BER of 1×10^{-5} . Furthermore, the proposed MU-2 protograph code provides at least a 0.4 dB gain over other protograph codes. In addition, the proposed MU-1 code, with a smaller base matrix, offers lower storage and decoding complexity. Although the performance of the proposed MU-1 protograph code is 0.15 dB worse than that of the proposed MU-2 code, it still outperforms other alternatives. These results confirm that multi-user codes designed based on the proposed principle possess a distinct advantage in SCMA systems.

VII. CONCLUSION

In this paper, we have investigated protograph-coded SCMA systems by introducing the high-performance R-SDD receiver. To address the inefficient information exchange in traditional IDD receivers, the R-SDD receiver dynamically adjusts the decoding order of users, focusing computational resources on worst-performing users. As a result, it achieves faster convergence and better performance with lower complexity compared with the traditional IDD receivers. To reduce the complexity of the MPA-based detector, an early-stopping criterion is conceived. Additionally, the proposed FG interleaving scheme further enhances system performance. Since traditional protographs are not well-suited for inter-user interference scenarios, we have also proposed a new principle for protograph code design tailored for SCMA systems. The resultant MU-2 protograph code outperforms existing competitors by over 0.3 dB in SCMA systems.

We conclude that the SCMA system with the proposed R-SDD receiver, FG interleaver and the newly designed protograph codes, is a promising solution for high-bandwidth, high-reliability communication systems, offering a new opportunity for 6G-enabled IoT applications.

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